CODER'S EVENT: TESSERACT

Problem Statement 4



TEAM ID: 46

TEAM NAME: TESSERACT

TEAM MEMBERS

1. SHWETA REKHA NATAK
2. KANISHKA MINA
3. ANIRUDHA KIRAN BHARADWAJ
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6. SAGUN SINGH

Department Inventory & Student Item Issue System

An automated inventory management and item issuing web application designed to manage department laboratory assets, equipment stock, student borrowing records, return deadlines, and renewal requests.

COORDINATOR MANAGEMENT

- **Inventory and Asset Tracking:** (Individual Assets like laptops, bulk items like cables)
- **Item Issuing and Validation:** (Student Registration, Condition, Stock, Asset Assignment)
- **Automatic Due Dates:** (Calculates 15-day borrowing period)
- **Returns and Partial Returns:** (Records quantities and condition: usable, damaged, lost, under-maintenance)
- **Renewal Management:** (Approve or Reject extension requests; extends due date by 15 days)
- **Reports and Exports:** (Department, Lab, Programme, Student, Item-wise; CSV export)

STUDENT PORTAL

- **Role-Based Access:** (View active items, due dates, renewal status, borrowing history)
- **Renewal Requests:** (Request extension before item due date)
- **Due-Date Indicators:** (Uses color-coded traffic-light indicators:
  🟢 Green — Upcoming or sufficiently distant due dates,
  🟡 Yellow — Due dates approaching,
  🔴 Red — Overdue items)

STUDENT EXPERIENCE & TOOLKIT

An automated inventory management and item issuing web application designed to manage department laboratory assets, equipment stock, student borrowing records, return deadlines, and renewal requests.

DASHBOARD VIEW

- View Active Issued Items
- Check Due Dates
- Monitor Renewal Status
- Review Borrowing History

RENEWAL REQUESTS

- Submit requests *before* an iten* an item is due
- Simple interface for date extensions



DUE DATE INDICATORS

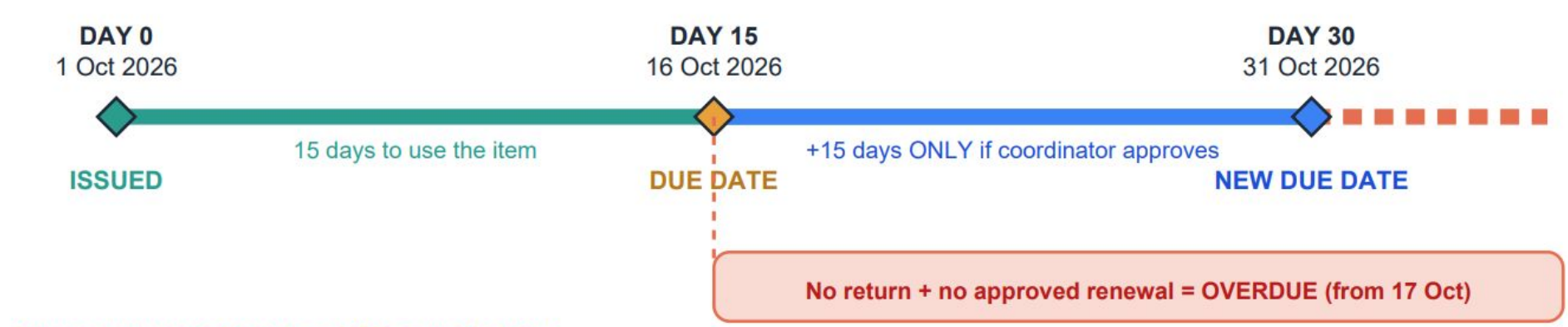


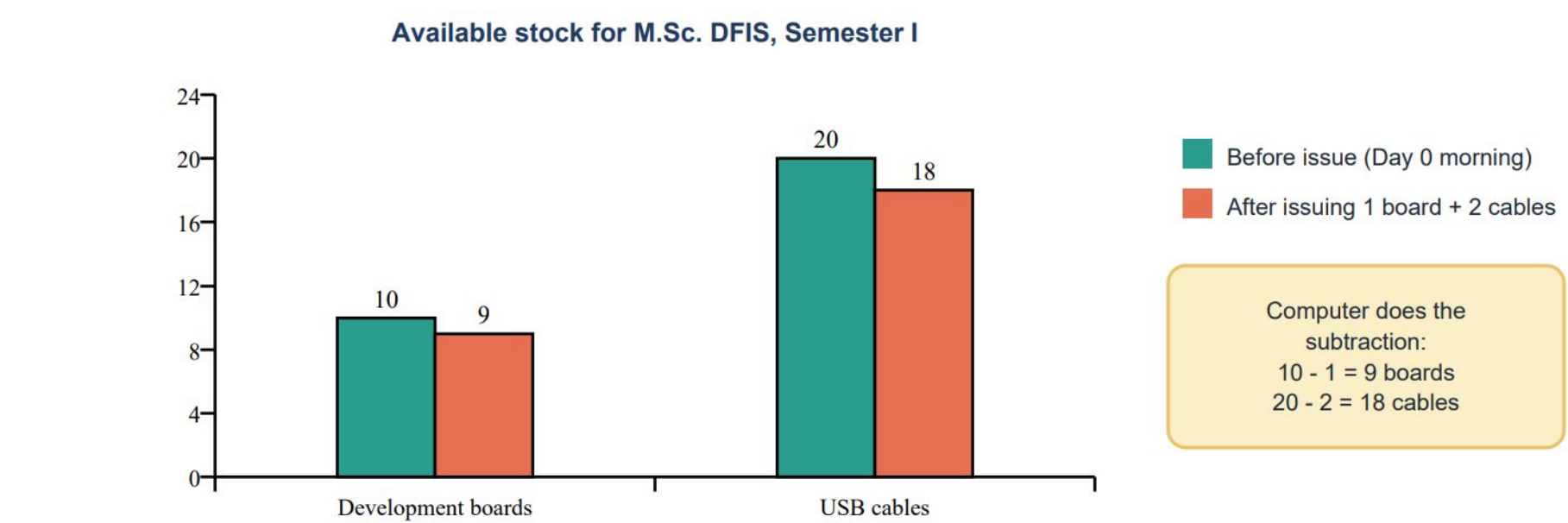
Upcoming (Distant)

Approaching (Soon)

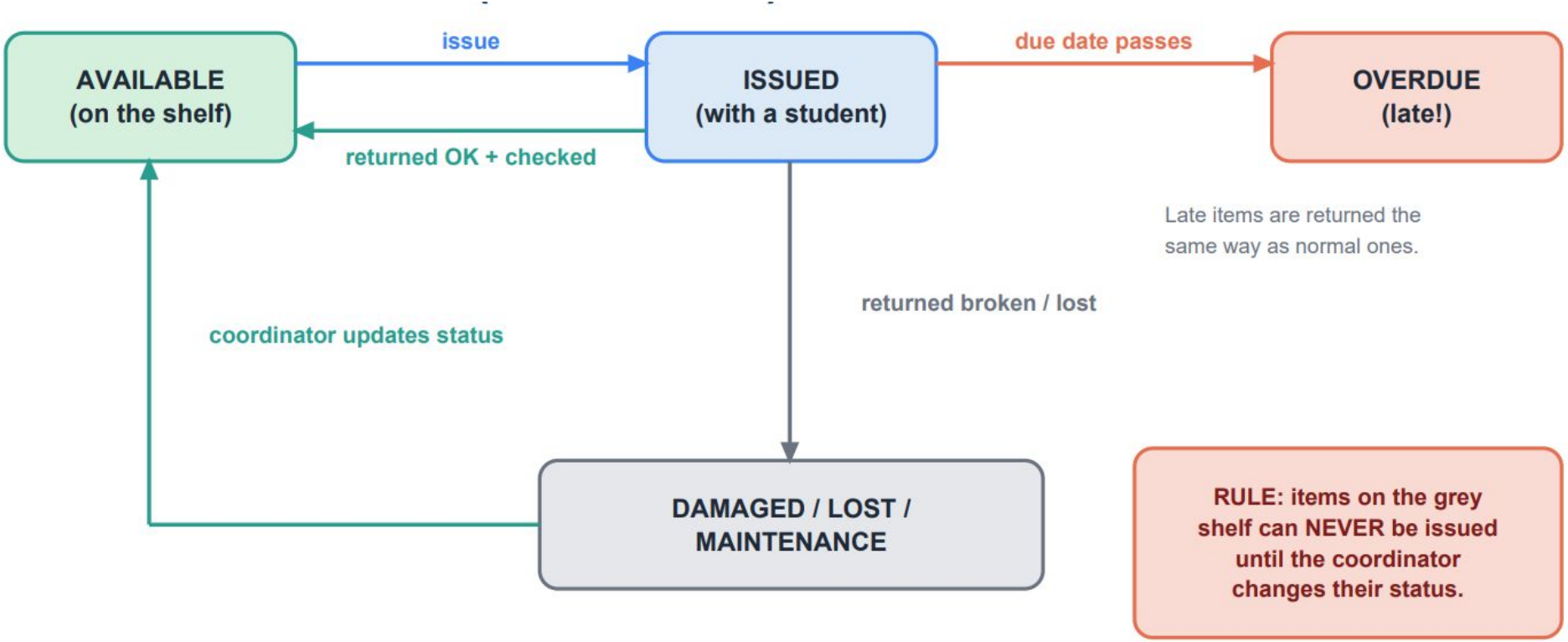
Overdue (ACTION)

- Traffic-light visual guidance for deadline awareness


DAY 0
1 Oct 2026
DAY 15
16 Oct 2026
DAY 30
31 Oct 2026
15 days to use the item
+15 days ONLY if coordinator approves
ISSUED
DUE DATE
NEW DUE DATE
No return + no approved renewal = OVERDUE (from 17 Oct)
Renewal request alone does NOT move the date!

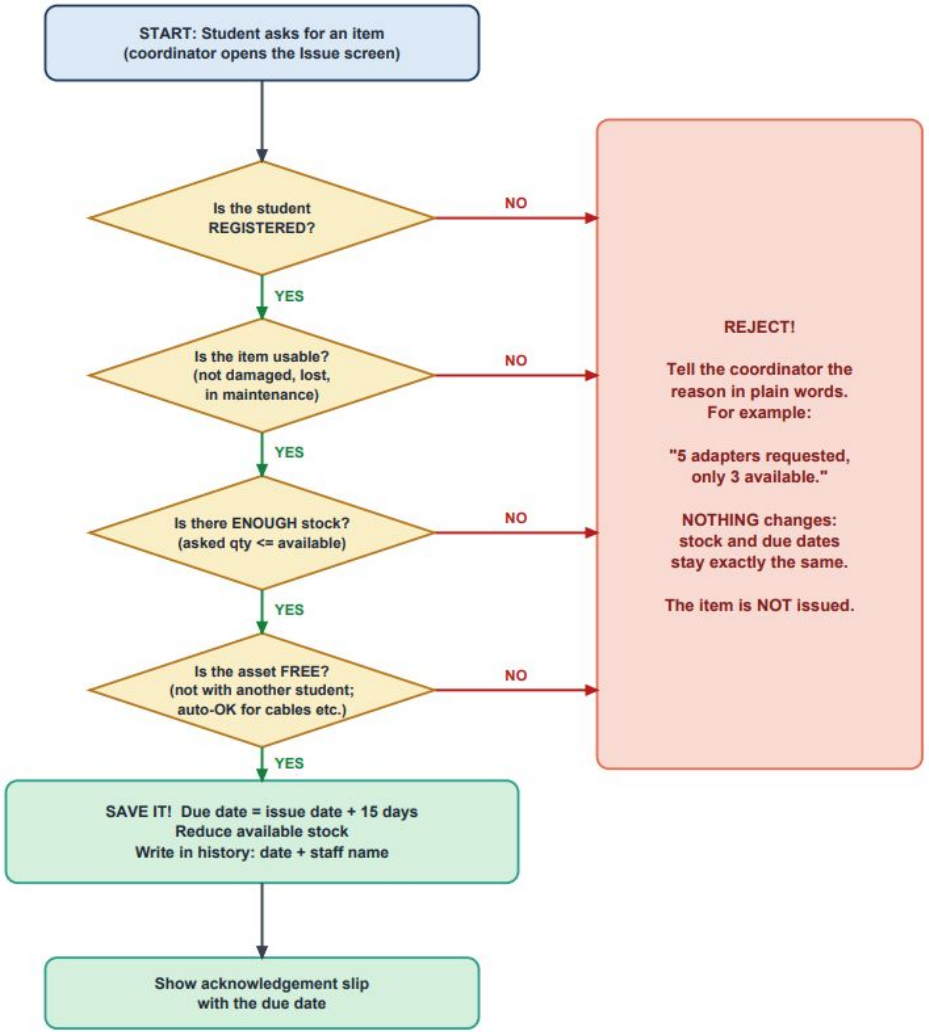

Available stock for M.Sc. DFIS, Semester I
24
20
16
12
8
4
0
10
9
20
18
Development boards
USB cables
Before issue (Day 0 morning)
After issuing 1 board + 2 cables
Computer does the subtraction:
10 - 1 = 9 boards
20 - 2 = 18 cables

STATUS CHART

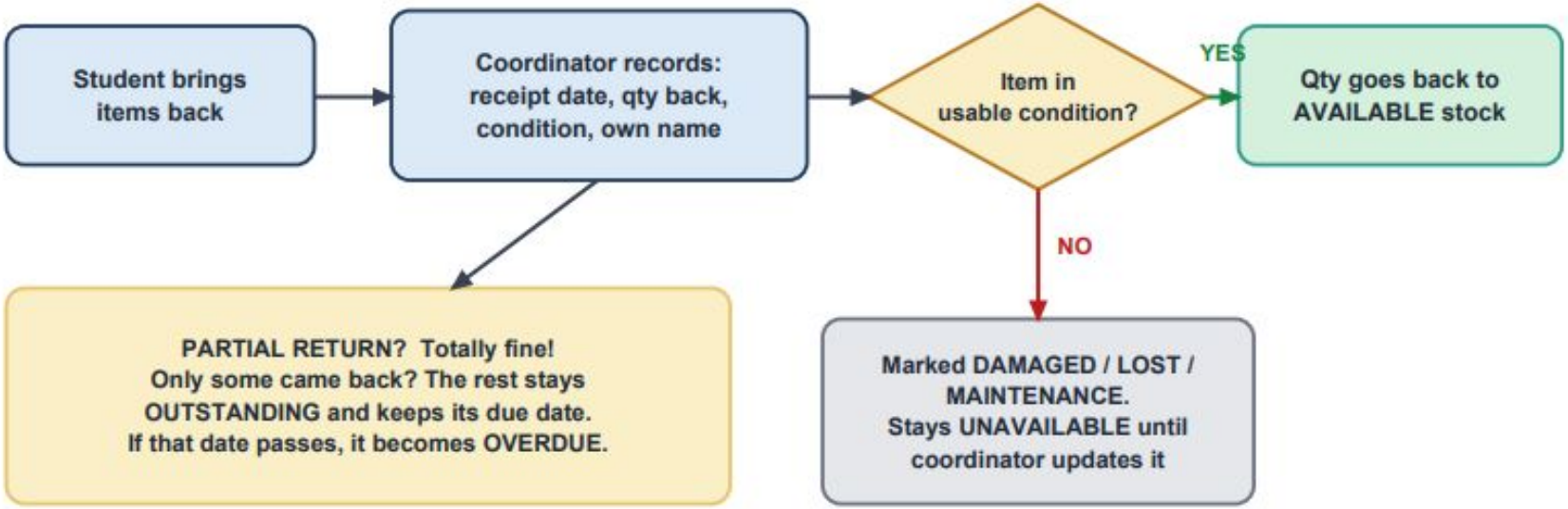


WORKFLOW

Issuing an item

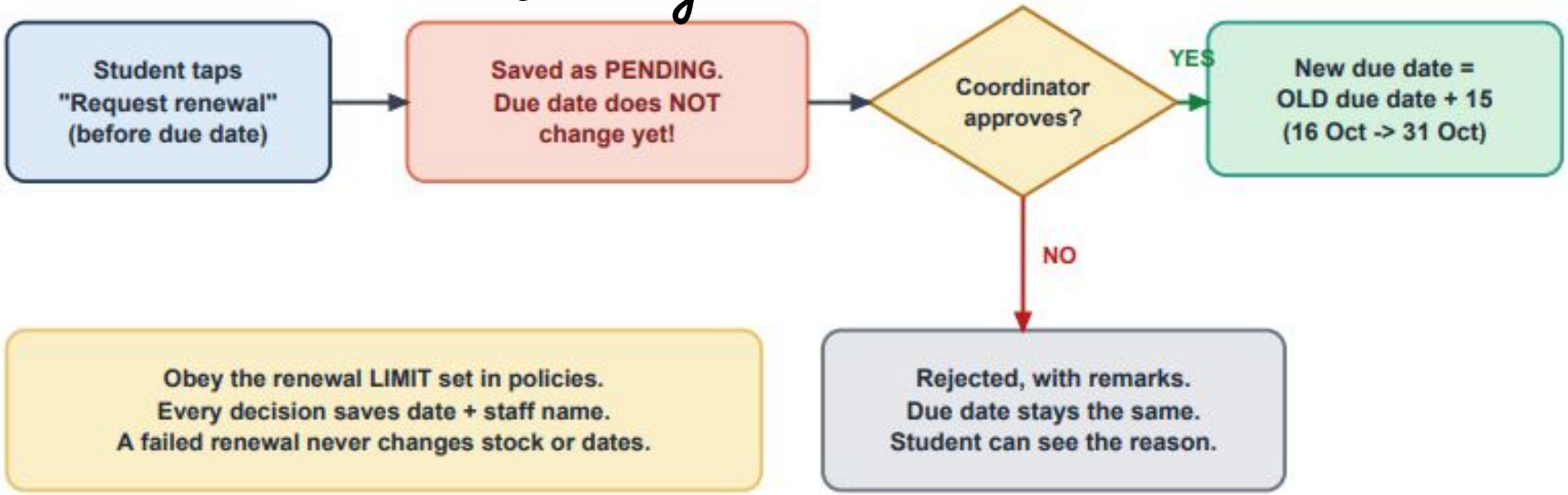


Returning an item



Safety rule: items return to the shelf ONLY AFTER the coordinator has recorded receipt AND condition.
No record = still counted as issued.

Renewing an item



Overdue Detector

Every time a page opens, the computer looks at all open issues: *"Is today later than the due date, and is anything still outstanding?"* If YES, it marks it **OVERDUE** and builds a sentence:
"Development Board DB-014 issued to student enrolment number XXXXX was due on 16 October 2026 and is overdue by 3 days."
Overdue days = today - due date. Items due soon (for example in the next 3 days) go into the **upcoming-due list**.

INNOVATION & TECHNOLOGY

KEY HIGHLIGHT FEATURES

BORROWER KARMA SCORE

- Starts at 100 Points
- Dynamically reflects borrower behavior
- Impacted by punctuality & item condition
- Badges: Gold, Silver, Bronze performance



DEMO TIME MACHINE

- Simulates future dates
- Demonstrates due-date calculations
- Highlights overdue item detection



TECHNOLOGY STACK

Python	Application Logic & Backend Processing
SQLite3	Database Management
HTML5	Web Page Structure